

Mortensen–Pissarides (1994)

Theorem and Proof Progress Audit

Lean4Tutorial / i002_replicate_MP_v2

28 July 2026

Purpose. This report distinguishes what the Lean development actually proves from equations that are merely defined, equilibrium conditions supplied as record fields, and economic conclusions already present in theorem hypotheses. The detailed source of truth is `proof_progress.md`; this PDF carries the same 79 result IDs, statuses, and recommendation in a print-oriented layout.

Lean source base	Lean4Tutorial/i002_replicate_MP
Paper	Mortensen-JobCreationJob-1994.pdf, printed pp. 397–415
Substantive results	79 across 16 theorem families
Formal headline	9 faithfully proved; 6 conditionally proved
Central blocker	No theorem identifies actual <code>ValueEquilibrium.S</code> with equation (12)
Recommended next result	Actual two-point surplus difference for <code>ValueEquilibrium.S</code>

Executive summary

- **Representation is substantial.** The main stationary equations, anticipated-state formulas, Markov Bellman equation, and accounting laws are expressible.
- **Derivation is concentrated in supporting algebra.** Value aggregation, zero profit, hazard/flow monotonicity, and steady-state accounting are the strongest current results.
- **The core paper chain is open.** Equation (12) is not proved for actual equilibrium surplus; equations (10) and (13) are not jointly derived from the value system; existence, uniqueness, and shock comparative statics are absent.
- **Quantitative replication has not started.** There is no concrete finite state, stochastic matrix, equilibrium witness, employment-distribution update, or simulation.

Print note. The enlarged right margin is intentional and reserved for handwritten review comments.

1 One-page high-level progress dashboard

Current formal status	Count	Share	Marker
Proved faithfully	9	9/79	PF
Proved conditionally	6	6/79	PC
Encoded as predicate/definition	8	8/79	ENC
Assumed in a structure	11	11/79	ASM
Elementary supporting lemma only	6	6/79	LEM
Finite-state witness only	0	0/79	FIN
Not formalized	33	33/79	NONE
Needs clarification/correction	6	6/79	CLAR

#	Theorem family / paper material	Represented	Derived	Adequacy
1	Primitives and stochastic environment, pp. 398–400	3/4	0/4	RED
2	Value equations and bargaining, (1)–(8)	4/4	3/4	AMBER
3	Surplus representation, (9),(12)	3/5	0/5	RED
4	Reservation productivity and JD, (9)–(10)	2/4	1/4	RED
5	Job creation, (12)–(13)	3/4	2/4	AMBER
6	Steady-state characterization, (10),(13),(14)	4/4	2/4	AMBER
7	Existence and uniqueness, p. 402 prose	0/3	0/3	RED
8	Comparative statics, pp. 401–404, Appendix	5/12	2/12	RED
9	Beveridge curve and unemployment, (14)–(15)	3/5	2/5	AMBER
10	Anticipated aggregate shocks, (16)–(30)	5/8	2/8	RED
11	Two-state dispersion shocks, (31)	0/3	0/3	RED
12	General Markov equilibrium, (32)–(34)	2/4	0/4	RED
13	Employment-distribution dynamics, (35)	1/3	0/3	RED
14	Simulation accounting, (36)–(38)	5/6	1/6	AMBER
15	Calibration and numerical replication	0/4	0/4	RED
16	Appendix derivations, (A1)–(A12)	0/6	0/6	RED

2 Status legend

Marker	Category	Meaning
PF	Proved faithfully	Lean derives the intended claim at appropriate generality.
PC	Proved conditionally	A sound proof retains an important supplied premise/restriction.
ENC	Encoded	Formula or object is a definition/predicate, not a derivation.
ASM	Assumed	The result is an equilibrium/path structure field.
LEM	Elementary lemma only	Detached algebra/order support, not the economic theorem.
FIN	Finite witness only	Concrete finite witness without a general theorem.
NONE	Not formalized	No substantive Lean result.
CLAR	Clarify/correct	Endpoint, assumptions, universality, or exactness must be repaired.

3 Theorem-level progress inventory

ID	State	Result / paper locus	Lean evidence and remaining gap
Family 1. Model primitives and stochastic environment			
PP-01-01	ENC	Affine job productivity p. 399: $p + \sigma\epsilon$.	Primitives.price, Definitions.lean: 64--66. Moment normalization is absent.
PP-01-02	NONE	Poisson iid standardized shock law pp. 399–400.	Raw fields at Definitions.lean: 25--40; no process, atomlessness, mean-zero, or variance-one contract.
PP-01-03	ENC	CRS matching with strict elasticities pp. 399–400.	mWorker, m, Definitions.lean: 52--62; only weak fields at lines 275–280.
PP-01-04	ENC	Threshold separation hazard p. 399: $\lambda F(\epsilon_d)$.	destructionHazard, Definitions.lean: 80--82; cutoff endpoint is inconsistent if the law has atoms.
Family 2. Value equations and Nash bargaining			
PP-02-01	ASM	Stationary value system (1)–(7) pp. 400–401.	Predicates Definitions.lean: 96--133; supplied fields Equilibrium.lean: 37--47.
PP-02-02	PC	Equations (3)–(7) imply (8) p. 401.	ValueEquilibrium.surplus_bellman, Results.lean: 24--38; paper mass/integrability closure is detached.
PP-02-03	PF	Firm value is zero at supplied cutoff p. 401.	ValueEquilibrium.J_cutoff_eq_zero, Results.lean: 40--47.
PP-02-04	PF	Free entry implies zero-profit value equation (1)–(2).	ValueEquilibrium.vacancy_free_entry_value, Results.lean: 49--62.

ID	State	Result / paper locus	Lean evidence and remaining gap
Family 3. Surplus representation			
PP-03-01	ENC	Closed-form cutoff surplus Equation (12).	S_cutoff, Definitions.lean:84--86; not identified with actual surplus.
PP-03-02	LEM	Closed-form surplus difference algebra Equation (12).	surplus_difference, Results.lean:74--79; applies only to the separate formula.
PP-03-03	NONE	Actual equilibrium surplus satisfies (12) p. 402.	No theorem for ValueEquilibrium.S; immediate critical blocker.
PP-03-04	ASM	Increasing surplus gives unique cutoff pp. 401–402.	cutoff_zero and reservation_rule are fields, Equilibrium.lean:48--49.
PP-03-05	NONE	Continuation expectation equals tail integral Equation (9).	Separate integrals at Definitions.lean:68--78; no layer-cake/Stieltjes bridge.
Family 4. Reservation productivity and job destruction			
PP-04-01	NONE	Integrated surplus equation (9) p. 401.	Requires actual (12) and tail identity.
PP-04-02	ASM	Job-destruction condition (10) pp. 401–402.	Predicate Definitions.lean:143--149; field Equilibrium.lean:64.
PP-04-03	PF	Higher cutoff raises hazard weakly p. 401.	destructionHazard_mono, Results.lean:109--114.
PP-04-04	CLAR	Cutoff endpoint matches destruction convention Paper assumes no atoms.	Lean uses Iic and weak cutoff acceptance; add atomlessness or use Iio.
Family 5. Job creation			
PP-05-01	ASM	Job-creation condition (13) p. 402.	Predicate Definitions.lean:159--164; field Equilibrium.lean:65.
PP-05-02	PC	Closed-form (13) iff closed-form zero profit Equation (13).	jobCreation_iff_freeEntryValue, Results.lean:121--145; actual J bridge missing.
PP-05-03	PC	JC locus slopes downward Figure 1 claim.	jobCreationCurve_slopes_down, Results.lean:147--188; supplied JC points.
PP-05-04	NONE	Every value equilibrium satisfies (13) Bridge claim.	Needs actual equation (12) plus zero-profit theorem.
Family 6. Steady-state characterization			
PP-06-01	ASM	Reduced equilibrium satisfies JD, JC, Beveridge (10),(13),(14).	SteadyStateEquilibrium, Equilibrium.lean:53--66.
PP-06-02	ENC	Vacancy, employment, and gross flows pp. 403–404.	Definitions Equilibrium.lean:72--83.
PP-06-03	PF	Stationary gross flows balance Equation (14).	gross_flows_balance, Results.lean:337--351.

ID	State	Result / paper locus	Lean evidence and remaining gap
PP-06-04	PF	Stationary unemployment drift is zero Equations (14)–(15).	unemploymentDrift_eq_zero, Results.lean:353--360.
Family 7. Existence and uniqueness			
PP-07-01	NONE	A steady state exists p. 402 prose.	No nonemptiness theorem; residual domains and boundary crossing missing.
PP-07-02	CLAR	The joint steady state is unique p. 402 prose.	Global conditions are unstated; prove sufficient-condition or local uniqueness theorem.
PP-07-03	NONE	Value equilibrium induces reduced equilibrium Central bridge.	No theorem from Equilibrium.lean:18--49 to lines 53–66.
Family 8. Comparative statics			
PP-08-01	NONE	Higher p raises tightness, lowers cutoff pp. 402–404.	Orderings are fields/hypotheses, not derived.
PP-08-02	PC	Given p-shock orders, flows move oppositely p. 404.	aggregateShock_flow_orders, Results.lean:307--317.
PP-08-03	LEM	Strict aggregate negative comovement p. 404.	Results.lean:378--386; strict flow orders are hypotheses.
PP-08-04	NONE	Higher b lowers tightness, raises cutoff pp. 401–404.	No paired-equilibrium theorem.
PP-08-05	NONE	Higher λ lowers cutoff and tightness Appendix (A1)–(A4).	No derivative/comparison theorem.
PP-08-06	NONE	Total destruction response to λ may reverse p. 404.	No total derivative decomposition.
PP-08-07	NONE	Higher r lowers tightness Appendix (A5)–(A7).	No theorem.
PP-08-08	CLAR	Cutoff response to r is ambiguous Appendix (A8).	Needs formula plus conditional signs or both-sign witnesses.
PP-08-09	LEM	Fixed-tightness dispersion RHS is positive Equation (11).	dispersionCutoffResponse_pos, Results.lean:190--207; no derivative equality.
PP-08-10	NONE	Higher σ raises equilibrium tightness/cutoff Appendix (A9)–(A12).	Moments, elasticity, and joint equilibrium derivative missing.
PP-08-11	PC	Given dispersion orders, both flows rise p. 404.	dispersionShock_flow_orders, Results.lean:319--329.
PP-08-12	LEM	Strict dispersion positive comovement p. 404.	Results.lean:388--396; strict flow orders are hypotheses.

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Family 9. Beveridge curve and unemployment			
PP-09-01	ASM	Beveridge formula (14) p. 403.	Predicate Definitions.lean:214--218; field Equilibrium.lean:66.
PP-09-02	PF	Beveridge denominator is positive Equation (14).	beveridge_denominator_pos, Results.lean:284--289.
PP-09-03	NONE	Zero drift iff equation (14) Equations (14)–(15).	Only the (14) to zero-drift direction is proved.
PP-09-04	CLAR	Beveridge curve slopes down under dominance p. 403 prose.	Paper's dominance condition must be stated as an exact inequality.
PP-09-05	PF	Unemployment rises iff destruction exceeds creation Equation (15).	unemploymentDrift_pos_iff, Results.lean:398--405.
Family 10. Two-state anticipated aggregate shocks			
PP-10-01	NONE	Two-state value equations (16)–(18) p. 405.	Only general Markov Bellman field exists; regional specialization missing.
PP-10-02	ASM	Anticipated destruction conditions (20),(24) p. 406.	Predicates Definitions.lean:166--193; fields Equilibrium.lean:124--127.
PP-10-03	ENC	Boom surplus formula (29) p. 407.	boomSurplus, Definitions.lean:195--201; no state-surplus equality.
PP-10-04	ASM	Boom job-creation condition (30) p. 407.	Predicate Definitions.lean:203--212; field Equilibrium.lean:131--132.
PP-10-05	PF	Zero transition reduces (30) to (13) p. 407.	boomJobCreation_zero_transition_iff, Results.lean:209--220.
PP-10-06	PC	Anticipation lowers boom tightness at fixed cutoffs p. 407.	anticipation_reduces_boom_tightness, Results.lean:222--282.
PP-10-07	NONE	Anticipation narrows cutoff gap/cyclical pp. 406–408.	Endogenous two-state response missing.
PP-10-08	NONE	Downturn destruction is dynamically more abrupt p. 408.	Only detached arithmetic at Results.lean:414--427.
Family 11. Two-state dispersion shocks			
PP-11-01	NONE	Anticipated dispersion-state equilibrium exists pp. 408–409.	No dispersion-specific state system.
PP-11-02	NONE	Direct creation effect dominates, equation (31) p. 409.	No theorem.
PP-11-03	NONE	Dispersion transitions yield same-direction flows/asymmetry pp. 408–409.	Only generic conditional flow lemma exists.

ID	State	Result / paper locus	Lean evidence and remaining gap
Family 12. General Markov equilibrium			
PP-12-01	ASM	Equations (32)–(34) characterize a Markov value equilibrium p. 409.	Fields of <code>GeneralMarkovValueEquilibrium</code> , <code>Equilibrium.lean:146--176</code> .
PP-12-02	ENC	General-state reduced equilibrium and kernel p. 409.	<code>MarkovEquilibrium</code> , <code>Equilibrium.lean:136--142</code> ; does not impose (32)–(34).
PP-12-03	NONE	A Markov equilibrium exists p. 409 prose.	No finite or general solution theorem.
PP-12-04	CLAR	Full-support max integral equals truncated paper integral Equation (34).	Statewise reservation/support closure missing from <code>Equilibrium.lean:156--176</code> .
Family 13. Employment-distribution dynamics			
PP-13-01	NONE	Employment measure has mass equal to employment p. 410.	No distribution state.
PP-13-02	NONE	Employment distribution obeys equation (35) p. 410.	No update operator or density theorem.
PP-13-03	ENC	One-off scrapping integral is expressible Equation (37).	<code>immediateScrapping</code> , <code>Definitions.lean:236--239</code> ; arbitrary integrand/path.
Family 14. Simulation accounting			
PP-14-01	ASM	Period creation obeys (36) p. 410.	Definition/field <code>Definitions.lean:232--234</code> , <code>Equilibrium.lean:194--195</code> .
PP-14-02	ASM	Period destruction obeys full (37) p. 410.	Additive field <code>Equilibrium.lean:196--197</code> ; ongoing term arbitrary.
PP-14-03	PF	Employment change equals creation minus destruction Equation (38).	<code>Results.lean:407--436</code> .
PP-14-04	LEM	Positive immediate term exceeds ongoing destruction Downturn prose.	<code>downturn_destruction_exceeds_ongoing</code> , <code>Results.lean:414--421</code> ; positivity supplied.
PP-14-05	LEM	Boom has no immediate term Boom prose.	<code>boom_without_scrapping</code> , <code>Results.lean:423--427</code> ; zero inserted.
PP-14-06	NONE	Simulation paths remain feasible Economic domain.	No bounds/nonnegativity in <code>Equilibrium.lean:181--199</code> .
Family 15. Calibration and numerical replication			
PP-15-01	NONE	Matrix (39) is stochastic p. 411.	No concrete <code>Fin3</code> matrix.
PP-15-02	NONE	Matching calibration satisfies (40) p. 411.	No calibrated matching function.

ID	State	Result / paper locus	Lean evidence and remaining gap
PP-15-03	NONE	Reported vectors (42) solve (32)–(34) p. 411.	No finite witness; rounded values require residual/enclosure certification.
PP-15-04	NONE	Simulation reproduces Table II p. 412.	No solver, RNG, statistics, or data reproduction.
Family 16. Appendix comparative-static derivations			
PP-16-01	NONE	A1–A3 give λ derivatives p. 413.	No differentiable residual system.
PP-16-02	NONE	Tail bound (A4) p. 414.	No interval-integral bound theorem.
PP-16-03	NONE	A5–A7 imply negative tightness response to r p. 414.	No theorem.
PP-16-04	CLAR	A8 gives ambiguous cutoff response to r p. 414.	Needs formula plus conditional signs or both-sign witnesses.
PP-16-05	NONE	A9–A10 give dispersion derivative system p. 414.	Current RHS definition is fixed-tightness only, <code>Definitions.lean:151--157</code> .
PP-16-06	NONE	A11–A12 imply dispersion orderings p. 414.	Mean-zero law, elasticity, and solved derivative signs missing.

4 What is actually proved today?

PF

Zero cutoff firm value; zero-profit vacancy value; weak hazard monotonicity; positive Beveridge denominator; gross-flow balance; zero drift; drift sign equivalence; zero-transition reduction of (30); employment accounting.

PC

Encoded (3)–(7) imply encoded (8); closed-form free-entry equivalence; JC slope; fixed-order aggregate and dispersion flow comparisons; fixed-cutoff anticipation comparison.

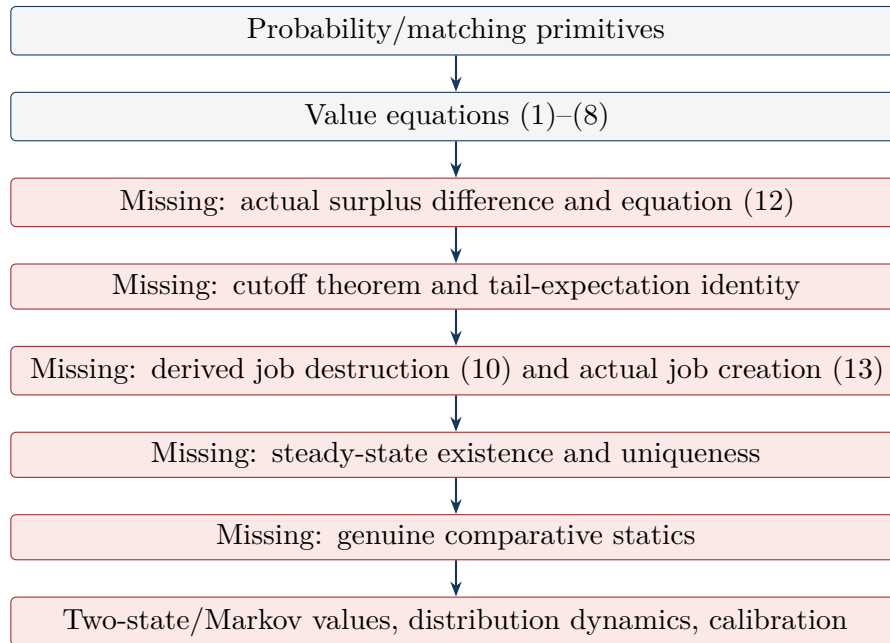
The exact declarations and locations appear in the inventory above. Each named theorem is covered by an `assert_no_sorry` entry at `Lean4Tutorial/i002_replicate_MP/Verification.lean:13--46`.

5 What is currently only assumed or represented?

- Equations (1)–(7), cutoff zero, and the reservation rule are `ValueEquilibrium` fields (`Equilibrium.lean:37--49`).
- Equations (10), (13), and (14) are `SteadyStateEquilibrium` fields (`Equilibrium.lean:64--66`).
- Boom/recession orderings are fields (`Equilibrium.lean:96--98`, `Equilibrium.lean:118--123`).
- Equations (20), (24), recession (13), and boom (30) are anticipated-equilibrium fields (`Equilibrium.lean:124--132`).

- Equations (32)–(34) are general Markov value-equilibrium fields (`Equilibrium.lean:161--176`).
- Equations (36)–(38) are path fields; ongoing destruction is arbitrary (`Equilibrium.lean:181--199`).
- `S_cutoff` and `boomSurplus` are formulas, not theorems about actual equilibrium surplus (`Definitions.lean:84--94`, `Definitions.lean:195--201`).

6 Dependency diagram and critical path



Three most important missing bridges

1. Actual `ValueEquilibrium.S` $\rightarrow$ equation (12) and `S_cutoff`.
2. Positive-continuation expectation $\rightarrow$ tail-CDF integral, enabling (9) and (10).
3. Derived JD/JC/Beveridge system $\rightarrow$ existence and uniqueness of a steady state.

7 First-milestone comparison

Candidate	Difficulty / risk	Immediate gain	Verdict
Probability/CDF foundation	Moderate-foundational; low circularity	Fixes atoms and enables measure/Appendix work	Essential second track, but no immediate headline theorem.
Strengthen derivation of (8)	Elementary-moderate; low risk	Cleans an already proved edge	Not the best new milestone.
Actual equation (12)	Elementary; low risk	First missing headline theorem; unlocks actual JC and cutoff work	GREEN Best first milestone.
Derive equation (10)	Foundational; high premature-assumption risk	Major JD bridge	Too early: tail identity and actual (12) missing.
Value-to-steady bridge	Foundational; high circularity risk	Closes architecture	End of first proof phase, not first step.

8 Recommended next proof milestone

Prove the two-point surplus-difference form of equation (12) for the actual ValueEquilibrium.S.

For A:AssumptionsP and E:ValueEquilibriumP, prove

$$E.S(\varepsilon_2) - E.S(\varepsilon_1) = \frac{\sigma(\varepsilon_2 - \varepsilon_1)}{r + \lambda}.$$

Use E.surplus_bellman at two shock values and subtract. Continuation and search terms cancel, so no new measure theory is required.

Why it dominates. It is upstream, elementary, noncircular, and establishes a fact about the actual value equilibrium rather than another parallel formula. It unlocks the actual equation (13) bridge, surplus monotonicity, and the next cutoff theorem. The probability/CDF foundation should follow before tail-integral or derivative work.

First three milestones

1. **Actual equation (12):** two-point surplus difference, then equality with S_cutoff. Difficulty: elementary. Circularity risk: low.
2. **Actual equation (13):** combine milestone 1 with zero profit and Nash sharing. Difficulty: elementary-moderate. Circularity risk: low.
3. **Paper probability/CDF foundation:** probability instance, atomlessness, moments, and derived CDF facts. Difficulty: moderate-foundational. Circularity risk: low.